	Marwadi University **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing(01CT1513)	**Aim:** Enhancement of Degraded Images Using Gray-Level Transformation Techniques	
Open Ended Problem: 07	**Date: 16/08/2026**	**Enrollment No: 92400133128**

Aim

To enhance five different degraded grayscale images using suitable gray-level transformation techniques and analyse the improvement in visual quality by comparing the original and enhanced images.
The experiment focuses on selecting appropriate intensity transformations according to the degradation present in each image and studying their effectiveness and limitations. This directly addresses the assignment requirement to enhance each of the five degraded images and document the observations for every enhancement step.

OBJECTIVES

1. To understand the concept of gray-level transformations in digital image processing.
2. To load and convert degraded images into grayscale format.
3. To analyze the intensity distribution of each degraded image.
4. To select suitable gray-level transformation techniques for different types of degradation.
5. To implement image negative transformation.
6. To implement logarithmic transformation.
7. To implement power-law/gamma transformation.
8. To implement contrast stretching.
9. To implement piecewise-linear gray-level transformation where appropriate.
10. To compare the original and enhanced images.
11. To observe how different transformations affect brightness and contrast.
12. To determine which transformation gives the best visual improvement for each degraded image.
13. To understand the limitations of gray-level transformations.
14. To implement the complete experiment using Python, NumPy, OpenCV, and Matplotlib.

INTRODUCTION

Digital images may suffer from poor contrast, insufficient brightness, excessive darkness, or limited intensity variation. Such degradation can make important features difficult to observe.
Gray-level transformation is one of the fundamental techniques used in Digital Image Processing to modify the intensity values of pixels in an image.
A gray-level transformation can generally be represented as:

$$s = T(r)$$

where:

- r = input pixel intensity
- s = output pixel intensity
- T = transformation function

Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing(01CT1513)	**Aim:** Enhancement of Degraded Images Using Gray-Level Transformation Techniques	
Open Ended Problem: 07	**Date: 16/08/2026**	**Enrollment No: 92400133128**

For an 8-bit grayscale image:

$$0 \leq r \leq 255$$

and normally:

$$0 \leq s \leq 255$$

The purpose of the transformation is to map the original intensity values into a more useful range so that hidden details become more visible.

THEORY

A grayscale image contains only intensity information rather than separate RGB color components.
For an 8-bit grayscale image:

0 = black
255 = white

Intermediate values represent different shades of gray.

GRAY-LEVEL TRANSFORMATION TECHNIQUES

For this experiment, the techniques can be selected according to the visible degradation of each image.

1) Image Negative:

The negative transformation is:

$$s = L - 1 - r$$

For an 8-bit image:

$$s = 255 - r$$

Dark pixels become bright and bright pixels become dark.
Suitable for Images where important details are more visible in the negative representation.

2) Logarithmic Transformation

The logarithmic transformation is:

$$s = c \cdot \log(1+r)$$

where:

$$c = 255 / \log(1+r_{max})$$

It expands low-intensity values and compresses high-intensity values.

Suitable for Images containing important details in dark regions.

Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing(01CT1513)	**Aim:** Enhancement of Degraded Images Using Gray-Level Transformation Techniques	
Open Ended Problem: 07	**Date: 16/08/2026**	**Enrollment No: 92400133128**

3) Power-Law / Gamma Transformation

The transformation is:

$$s = c.r \wedge \gamma$$

A normalized implementation is:

$$s = 255 (r / 255) \wedge \gamma$$

For $\gamma < 1$: dark regions are brightened.
For $\gamma > 1$: the image becomes darker.
Suitable for Dark or underexposed images.

4) Contrast Stretching

Contrast stretching expands the range of pixel intensities.
A simple transformation is:

$$s = (r – rmin / rmax – rmin) * 255$$

This increases the separation between dark and bright regions.
Suitable for Low-contrast images.

5) Piecewise Linear Transformation

Different intensity ranges can be mapped independently.
For example:

$$s = \begin{cases} 0, & r < r_1 \\ \frac{r - r_1}{r_2 - r_1}(s_2 - s_1) + s_1, & r_1 \le r \le r_2 \\ 255, & r > r_2 \end{cases}$$

This allows selected intensity ranges to be enhanced more strongly.

Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing(01CT1513)	**Aim:** Enhancement of Degraded Images Using Gray-Level Transformation Techniques	
Open Ended Problem: 07	**Date: 16/08/2026**	**Enrollment No: 92400133128**

Image 1

Objective

To enhance the contrast of the given low-contrast grayscale image using suitable gray-level transformation techniques, so that the details of the main building and the surrounding background become clearer and more distinguishable.

Theory

The given image is a low-contrast grayscale image in which the intensity values are concentrated within a relatively narrow range. Due to the low difference between the dark and bright regions, the details of the building, trees, and background are not clearly visible.
To improve the visual quality of the image, **contrast stretching** can be applied. Contrast stretching expands the existing range of gray-level intensities to a wider range, usually from 0 to 255. This increases the difference between dark and bright regions and makes important features more visible.
The contrast-stretching transformation can be represented as:
$s = (r - rmin) / (rmax - rmin) \times 255$
where:

- **r** = input gray-level intensity
- **s** = output gray-level intensity
- **rmin** = minimum intensity value in the input image
- **rmax** = maximum intensity value in the input image

By expanding the intensity range, the contrast of the image is improved and the architectural details of the main building become more prominent.

Additional methods

Gamma transformation:

Gamma transformation can also be used to improve the brightness and visibility of image regions.
$s = 255(r / 255)^{\gamma}$
For **$\gamma < 1$**, darker regions are brightened, which can help reveal details that are difficult to observe in the original image.

2. Log Transformation
Log transformation enhances details in darker intensity regions.
$s = c \log(1 + r)$
This transformation expands lower gray-level values and can make hidden details in dark areas of the image more visible.

Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing(01CT1513)	**Aim:** Enhancement of Degraded Images Using Gray-Level Transformation Techniques	
Open Ended Problem: 07	**Date: 16/08/2026**	**Enrollment No: 92400133128**

Suitable Methods

1. Original grayscale image
2. Contrast stretching
3. Gamma transformation
4. Logarithmic transformation

Primary method: Contrast stretching.

Algorithm

 Read the given low-contrast image.
 Convert the image to grayscale.
 Determine the minimum and maximum pixel intensity values.
 Apply contrast stretching to expand the gray-level range.
 Apply gamma transformation to improve the visibility of dark regions.
 Apply logarithmic transformation to enhance details in low-intensity areas.
 Normalize the enhanced image to the range 0–255.
 Display the original and enhanced images.
 Compare the visual quality and contrast of the results.
 Select the most effective transformation for improving the image.

Display Image 1

Contrast Stretching

Code:

Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing(01CT1513)	**Aim:** Enhancement of Degraded Images Using Gray-Level Transformation Techniques	
Open Ended Problem: 07	**Date: 16/08/2026**	**Enrollment No: 92400133128**

```python
import cv2
import numpy as np
import matplotlib.pyplot as plt

# Read Image 1
src_image = cv2.imread('/content/Image1_LowContrast.jpg',
                       cv2.IMREAD_GRAYSCALE)

# Check whether image is loaded
if src_image is None:
    print("Image not found!")
else:

    # -------------------------
    # Contrast Stretching
    # -------------------------
    min_val = np.min(src_image)
    max_val = np.max(src_image)

    enhanced_image = ((src_image - min_val) /
                      (max_val - min_val) * 255).astype(np.uint8)

    # -------------------------
    # Display Original & Enhanced
    # -------------------------
    plt.figure(figsize=(12, 5))

    plt.subplot(1, 2, 1)
    plt.imshow(src_image, cmap='gray')
    plt.title("Original - Low Contrast")
    plt.axis('off')

    plt.subplot(1, 2, 2)
    plt.imshow(enhanced_image, cmap='gray')
    plt.title("Enhanced - Contrast Stretching")
    plt.axis('off')

    plt.tight_layout()
    plt.show()
```

Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing(01CT1513)	**Aim:** Enhancement of Degraded Images Using Gray-Level Transformation Techniques	
Open Ended Problem: 07	**Date: 16/08/2026**	**Enrollment No: 92400133128**

Output:



Observation – Image 1

- The original image appears hazy and has low overall contrast.
- The details of the main building are not clearly separated from the surrounding background.
- Contrast stretching improves the separation between dark and bright regions.
- Gamma transformation increases the visibility of darker portions of the image.
- Logarithmic transformation enhances information present in the lower intensity regions.
- Among the applied methods, contrast stretching gives the most noticeable improvement in overall image contrast.

Conclusion – Image 1

The given low-contrast image was successfully enhanced using different gray-level transformation techniques. **Contrast stretching is the most effective method for this image** because it expands the limited intensity range and improves the visibility of the building and surrounding details. Gamma and logarithmic transformations provide additional enhancement, especially in the darker regions.

Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing(01CT1513)	**Aim:** Enhancement of Degraded Images Using Gray-Level Transformation Techniques	
Open Ended Problem: 07	**Date: 16/08/2026**	**Enrollment No: 92400133128**

IMAGE 2

Objective

To enhance the visibility of the given underexposed grayscale image using suitable gray-level transformation techniques, especially to brighten the dark regions and reveal details of the trees, pathway, and surrounding forest.

Theory

The given image is an **underexposed grayscale image**, where most of the pixel intensity values are concentrated in the lower gray-level range. As a result, the trees, leaves, and pathway appear very dark and many details are difficult to distinguish.

A suitable technique for this image is **Gamma transformation**. Gamma transformation is useful for adjusting the brightness of an image. When $\gamma < 1$, darker regions are expanded and brightened, making details in the shadow areas more visible.

The gamma transformation equation is:

$$s = 255(r / 255)^{\gamma}$$

where:

- **r** = input gray-level intensity
- **s** = output gray-level intensity
- γ = gamma value

For $\gamma < 1$, the dark regions of the image are brightened while preserving the overall structure of the image.

Additional Methods

1. Contrast Stretching

Contrast stretching expands the range of gray-level intensities and increases the difference between dark and bright regions. It can improve the overall contrast of the forest image.

2. Logarithmic Transformation

Log transformation expands lower gray-level values and is useful for revealing details hidden in dark regions.

$s = c \log(1 + r)$

Since the main problem in this image is low brightness, gamma transformation with $\gamma < 1$ is expected to provide the most effective enhancement.

	Marwadi University **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing(01CT1513)	**Aim:** Enhancement of Degraded Images Using Gray-Level Transformation Techniques	
Open Ended Problem: 07	**Date: 16/08/2026**	**Enrollment No: 92400133128**

Suitable Methods

1. Original grayscale
2. Log transformation
3. Gamma transformation
4. Contrast stretching

Primary method: Gamma transformation ($\gamma < 1$).

Algorithm

1. Read the given underexposed image.
2. Convert the image to grayscale.
3. Analyze the intensity range of the image.
4. Apply logarithmic transformation.
5. Apply gamma transformation with $\gamma < 1$.
6. Apply contrast stretching.
7. Normalize the output intensity values to **0–255**.
8. Display the original and enhanced images.
9. Compare the visibility of dark regions and image details.
10. Select **Gamma transformation** as the most effective enhancement method.

Original Image:

Code:

Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing(01CT1513)	**Aim:** Enhancement of Degraded Images Using Gray-Level Transformation Techniques	
Open Ended Problem: 07	**Date: 16/08/2026**	**Enrollment No: 92400133128**

```python
import cv2
import numpy as np
import matplotlib.pyplot as plt

# Read Image 2
src_image = cv2.imread('/content/Image2_Underexposed.jpg',
                       cv2.IMREAD_GRAYSCALE)

# Check whether image is loaded
if src_image is None:
    print("Image not found!")
else:

    # -------------------------
    # Gamma Correction
    # -------------------------
    gamma = 2.5

    normalized_image = src_image / 255.0

    gamma_corrected = np.power(normalized_image, 1 / gamma)

    enhanced_image = np.uint8(gamma_corrected * 255)

    # -------------------------
    # Display Images
    # -------------------------
    plt.figure(figsize=(12, 5))

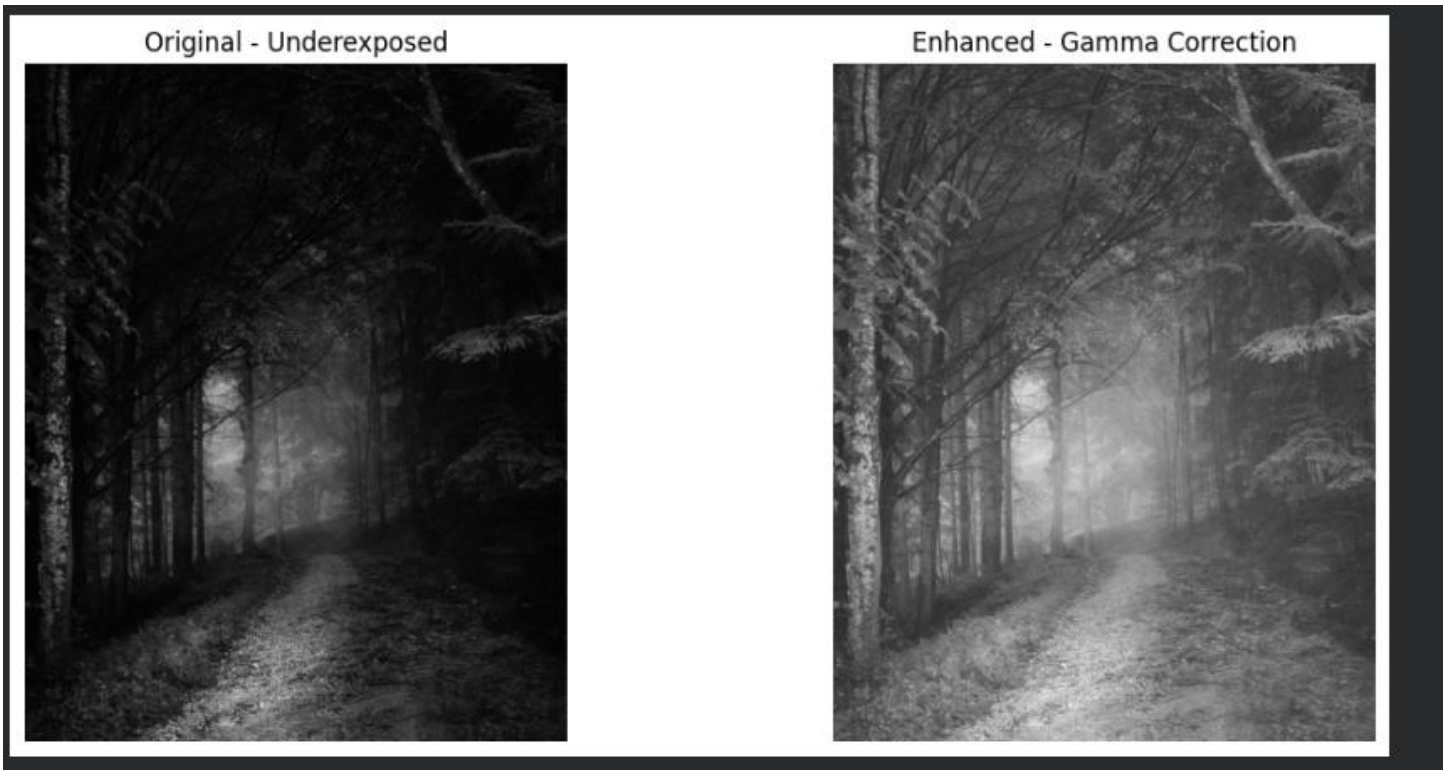
    plt.subplot(1, 2, 1)
    plt.imshow(src_image, cmap='gray', vmin=0, vmax=255)
    plt.title("Original - Underexposed")
    plt.axis('off')

    plt.subplot(1, 2, 2)
    plt.imshow(enhanced_image, cmap='gray', vmin=0, vmax=255)
    plt.title("Enhanced - Gamma Correction")
    plt.axis('off')

    plt.tight_layout()
    plt.show()
```

Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing(01CT1513)	**Aim:** Enhancement of Degraded Images Using Gray-Level Transformation Techniques	
Open Ended Problem: 07	**Date: 16/08/2026**	**Enrollment No: 92400133128**

Output:



Observation – Image 2

- The original image is underexposed, with most regions appearing very dark.
- Important details of the trees, leaves, and pathway are difficult to observe in the original image.
- Gamma correction with $\gamma < 1$ significantly brightens the darker regions.
- Details in the forest and pathway become more visible after gamma correction.
- The enhanced image provides better visibility of objects in the shadow regions.
- Gamma correction gives a clear improvement in brightness and visibility for this image.

Conclusion – Image 2

The given underexposed image was successfully enhanced using gray-level transformation techniques. **Gamma correction with $\gamma < 1$ is the most suitable technique for this image** because it effectively brightens the dark regions while revealing details of the trees and pathway. The enhanced image has better overall visibility compared with the original underexposed image.

Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing(01CT1513)	**Aim:** Enhancement of Degraded Images Using Gray-Level Transformation Techniques	
Open Ended Problem: 07	**Date: 16/08/2026**	**Enrollment No: 92400133128**

IMAGE 3

Objective

To enhance the given overexposed image using suitable gray-level transformation techniques, with the aim of reducing excessive brightness and improving the visibility of details in the buildings, road, vehicles, and surrounding areas.

Theory

The given image is an **overexposed image**, where a large portion of the pixel intensity values is concentrated toward the higher end of the gray-level range. Due to excessive brightness, several regions appear washed out and important details are difficult to distinguish.

A suitable technique for this image is **Gamma transformation with $\gamma > 1$**. Gamma transformation modifies the intensity values of an image and can be used to reduce excessive brightness.

The gamma transformation equation is:

$$s = 255(r / 255)^{\gamma}$$

where:

- **r** = input gray-level intensity
- **s** = output gray-level intensity
- **γ** = gamma value

For **$\gamma > 1$**, brighter intensity values are compressed, which helps reduce excessive brightness and makes some details more distinguishable.

Additional Methods

1. Contrast Stretching

Contrast stretching adjusts the intensity range of the image to improve the difference between different gray levels. It can help enhance the remaining visible details.

2. Logarithmic Transformation

Logarithmic transformation expands lower intensity values and compresses higher intensity values. It can be used for comparison, although it may not be as suitable as gamma correction for a strongly overexposed image.

Therefore, **Gamma transformation with $\gamma > 1$ is considered the primary method** for enhancing this image because the main problem is excessive brightness.

Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing(01CT1513)	**Aim:** Enhancement of Degraded Images Using Gray-Level Transformation Techniques	
Open Ended Problem: 07	**Date: 16/08/2026**	**Enrollment No: 92400133128**

Suitable Methods

1. Original grayscale
2. Gamma transformation
3. Log transformation
4. Contrast stretching

Primary method: Gamma Transformation ($\gamma > 1$).

Algorithm

- Read the given overexposed image.
- Convert the image to grayscale.
- Analyze the intensity range of the image.
- Apply gamma transformation with **$\gamma > 1$** to reduce excessive brightness.
- Apply logarithmic transformation for comparison.
- Apply contrast stretching to improve the remaining intensity range.
- Normalize the output intensity values to **0–255**.
- Display the original and enhanced images.
- Compare the visibility of buildings, road, vehicles, and other details.
- Select **Gamma transformation ($\gamma > 1$)** as the most effective enhancement method.

Original Image

Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing(01CT1513)	**Aim:** Enhancement of Degraded Images Using Gray-Level Transformation Techniques	
Open Ended Problem: 07	**Date: 16/08/2026**	**Enrollment No: 92400133128**

Code:

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

src_image = cv2.imread('/content/Image3_Overexposed.png',
                       cv2.IMREAD_GRAYSCALE)

if src_image is None:
    print("Image not found!")
else:

    gamma = 2.5

    normalized_image = src_image / 255.0

    enhanced_image = np.power(normalized_image, gamma)

    enhanced_image = np.uint8(enhanced_image * 255)

    plt.figure(figsize=(12, 5))

    plt.subplot(1, 2, 1)
    plt.imshow(src_image, cmap='gray', vmin=0, vmax=255)
    plt.title("Original - Overexposed")
    plt.axis('off')

    plt.subplot(1, 2, 2)
    plt.imshow(enhanced_image, cmap='gray', vmin=0, vmax=255)
    plt.title("Enhanced - Gamma Correction")
    plt.axis('off')

    plt.tight_layout()
    plt.show()
```

Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing(01CT1513)	**Aim:** Enhancement of Degraded Images Using Gray-Level Transformation Techniques	
Open Ended Problem: 07	**Date: 16/08/2026**	**Enrollment No: 92400133128**

Output:



Observation

- The original image is highly overexposed, with large regions appearing extremely bright.
- Details in the sky, buildings, road, and background are partially washed out.
- Gamma transformation with $\gamma > 1$ reduces the excessive brightness of the image.
- The enhanced image provides better visibility of the buildings, trees, road, and vehicle.
- Darker details become more distinguishable after gamma correction.
- Gamma transformation gives the most noticeable improvement for this overexposed image..

Conclusion

The given overexposed image was successfully enhanced using gray-level transformation techniques. **Gamma transformation with $\gamma > 1$ is the most suitable technique** because it reduces excessive brightness and improves the visibility of important image details. The enhanced image provides better contrast and clearer structural information compared with the original image.

Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing(01CT1513)	**Aim:** Enhancement of Degraded Images Using Gray-Level Transformation Techniques	
Open Ended Problem: 07	**Date: 16/08/2026**	**Enrollment No: 92400133128**

IMAGE 4

Objective

To enhance the given foggy grayscale image using suitable gray-level transformation techniques, with the aim of improving contrast and making the trees, pathway, and background details more clearly visible.

Theory

he given image has **low contrast due to fog**, which causes the intensity values to be concentrated within a limited gray-level range. The fog reduces the difference between the foreground and background, making distant trees and other details appear faint.
Contrast stretching is a suitable technique for this image because it expands the limited range of pixel intensities over a wider range, usually from **0 to 255**. This increases the difference between dark and bright regions and improves the visibility of objects.
The contrast-stretching equation is:
s = (r − rmin) / (rmax − rmin) × 255
where:

- **r** = input gray-level intensity
- **s** = output gray-level intensity
- **rmin** = minimum input intensity
- **rmax** = maximum input intensity

Additional Methods

1. Gamma Transformation

Gamma transformation can adjust the brightness and contrast of the foggy regions. Different gamma values can be tested to improve visibility.

2. Logarithmic Transformation

Log transformation enhances lower-intensity information and can help reveal details in darker areas of the forest.
Therefore, Contrast Stretching is considered the primary method because the major problem in this image is reduced contrast caused by fog.

	Marwadi University **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing(01CT1513)	**Aim:** Enhancement of Degraded Images Using Gray-Level Transformation Techniques	
Open Ended Problem: 07	**Date: 16/08/2026**	**Enrollment No: 92400133128**

Suitable Methods

1. Original grayscale
2. Gamma transformation
3. Log transformation
4. Contrast stretching

Primary method: Histogram Equalization..

Algorithm

1. Read the given foggy image.
2. Convert the image to grayscale.
3. Analyze the minimum and maximum intensity values.
4. Apply contrast stretching to expand the intensity range.
5. Apply gamma transformation for comparison.
6. Apply logarithmic transformation.
7. Normalize the output to **0–255**.
8. Display the original and enhanced images.
9. Compare the visibility of trees, pathway, and background.
10. Select the most effective enhancement method.

Original Image

Code:

Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing(01CT1513)	**Aim:** Enhancement of Degraded Images Using Gray-Level Transformation Techniques	
Open Ended Problem: 07	**Date: 16/08/2026**	**Enrollment No: 92400133128**

```python
import cv2
import matplotlib.pyplot as plt

# Read Image 4
src_image = cv2.imread('/content/Image4_Foggy.jpg',
                       cv2.IMREAD_GRAYSCALE)

# Check whether image is loaded
if src_image is None:
    print("Image not found!")
else:

    # -------------------------
    # Histogram Equalization
    # -------------------------
    enhanced_image = cv2.equalizeHist(src_image)

    # -------------------------
    # Display Original & Enhanced
    # -------------------------
    plt.figure(figsize=(12, 5))

    plt.subplot(1, 2, 1)
    plt.imshow(src_image, cmap='gray', vmin=0, vmax=255)
    plt.title("Original - Foggy")
    plt.axis('off')

    plt.subplot(1, 2, 2)
    plt.imshow(enhanced_image, cmap='gray', vmin=0, vmax=255)
    plt.title("Enhanced - Histogram Equalization")
    plt.axis('off')

    plt.tight_layout()
    plt.show()
```

Output:

Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing(01CT1513)	**Aim:** Enhancement of Degraded Images Using Gray-Level Transformation Techniques	
Open Ended Problem: 07	**Date: 16/08/2026**	**Enrollment No: 92400133128**



Observation – Image 4

- The original image appears foggy and has low contrast.
- The trees and pathway in the background are not clearly distinguishable due to the haze.
- Histogram equalization improves the distribution of gray-level intensities.
- The enhanced image shows better separation between dark and bright regions.
- Details of the trees, ground, and pathway become more visible after enhancement.
- Histogram equalization provides a significant improvement in the overall contrast of the foggy image.

Conclusion – Image 4

The given foggy image was successfully enhanced using gray-level image processing techniques. **Histogram equalization is the most suitable method for this image** because it improves the contrast and makes the forest structure and pathway more clearly visible. The enhanced image provides better visual detail and separation compared with the original foggy image.

Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing(01CT1513)	**Aim:** Enhancement of Degraded Images Using Gray-Level Transformation Techniques	
Open Ended Problem: 07	**Date: 16/08/2026**	**Enrollment No: 92400133128**

Image 5

Objective

To enhance the given noisy grayscale image using suitable gray-level transformation techniques, with the aim of improving contrast and making important details such as the cabin, trees, mountains, and lake more clearly visible.

Theory

he given image contains **noise along with low contrast**, causing the scene to appear faded and reducing the visibility of important details. The intensity values are distributed within a limited range, resulting in poor separation between the dark and bright regions.
A suitable technique for this image is **contrast stretching**. It expands the existing range of gray-level intensities to a wider range, usually from **0 to 255**. This improves the difference between dark and bright areas and makes the objects in the image more distinguishable.
The contrast-stretching equation is:
$$s = (r - r_{min}) / (r_{max} - r_{min}) \times 255$$
where:

- **r** = input gray-level intensity
- **s** = output gray-level intensity
- **rmin** = minimum input intensity
- **rmax** = maximum input intensity

By expanding the intensity range, contrast stretching improves the overall appearance of the image and makes features such as the trees, cabin, mountains, rocks, and water more visible.

Additional Methods

1. Gamma Transformation
Gamma transformation can be used to adjust the brightness of different intensity regions and improve the visibility of details.
$$s = 255(r / 255)^{\gamma}$$

2. Logarithmic Transformation
Logarithmic transformation expands lower gray-level values and can enhance details in darker regions.
$$s = c \log(1 + r)$$
For this image, **contrast stretching is considered the primary method** because the main requirement is to improve the limited contrast and overall visibility of the noisy scene.

Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing(01CT1513)	**Aim:** Enhancement of Degraded Images Using Gray-Level Transformation Techniques	
Open Ended Problem: 07	**Date: 16/08/2026**	**Enrollment No: 92400133128**

Suitable Methods

1. Original grayscale
2. Gamma transformation
3. Log transformation
4. Contrast stretching

Primary method: Contrast stretching.

Algorithm

1. Read the given noisy image.
2. Convert the image to grayscale.
3. Analyze the intensity range of the image.
4. Apply contrast stretching to expand the gray-level range.
5. Apply gamma transformation for comparison.
6. Apply logarithmic transformation.
7. Normalize the output intensity values to **0–255**.
8. Display the original and enhanced images.
9. Compare the visibility of the cabin, trees, mountains, and lake.
10. Select **Contrast stretching** as the most effective enhancement method.

Original Image

Code:

Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing(01CT1513)	**Aim:** Enhancement of Degraded Images Using Gray-Level Transformation Techniques	
Open Ended Problem: 07	**Date: 16/08/2026**	**Enrollment No: 92400133128**

```python
import cv2
import numpy as np
import matplotlib.pyplot as plt

# Read Image 5
src_image = cv2.imread('/content/Image5_Noisy.png',
                       cv2.IMREAD_GRAYSCALE)

# Check whether image is loaded
if src_image is None:
    print("Image not found!")
else:

    # -------------------------
    # Contrast Stretching
    # -------------------------
    min_val = np.min(src_image)
    max_val = np.max(src_image)

    enhanced_image = ((src_image - min_val) /
                      (max_val - min_val) * 255).astype(np.uint8)

    # -------------------------
    # Display Original & Enhanced
    # -------------------------
    plt.figure(figsize=(12, 5))

    plt.subplot(1, 2, 1)
    plt.imshow(src_image, cmap='gray', vmin=0, vmax=255)
    plt.title("Original - Noisy")
    plt.axis('off')

    plt.subplot(1, 2, 2)
    plt.imshow(enhanced_image, cmap='gray', vmin=0, vmax=255)
    plt.title("Enhanced - Contrast Stretching")
    plt.axis('off')

    plt.tight_layout()
    plt.show()
```

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Output:



Observation – Image 5

- The original image appears faded and contains noticeable noise.
- The cabin, trees, mountains, rocks, and lake have limited contrast in the original image.
- Contrast stretching increases the difference between dark and bright regions.
- The enhanced image shows clearer details in the cabin, trees, mountains, and surrounding landscape.
- The overall visibility and sharpness of important features are improved after contrast stretching.
- Contrast stretching provides the most effective overall enhancement for this image.

Conclusion – Image 5

The given noisy image was successfully enhanced using gray-level transformation techniques. **Contrast stretching is the most suitable method for this image** because it expands the limited intensity range and improves the visibility of important features. The enhanced result provides better contrast and clearer structural details compared with the original image.

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LIMITATIONS

Gray-level transformations have several limitations:

1. They modify pixel intensities but do not create new information.
2. They cannot completely recover details lost during image acquisition.
3. Enhancement may also amplify noise.
4. Choosing unsuitable parameters can produce over-enhancement.
5. One transformation may work well for one image but poorly for another.
6. Gray-level transformations alone are not sufficient for all types of degradation.
7. Images with severe noise may require spatial filtering.
8. Images with blur may require restoration/deconvolution techniques.
9. Images with complex illumination problems may require more advanced methods.

The assignment specifically asks the final conclusion to address both the **effectiveness and limitations** of the gray-level techniques.

APPLICATIONS

Medical Imaging:

Gray-level enhancement can improve visibility of structures in X-rays, CT scans, MRI images, and other medical images.

Satellite Image Processing:

Low-contrast satellite images can be enhanced to improve the visibility of terrain and environmental features.

Industrial Inspection:

Enhancement can help identify defects, cracks, and irregularities in manufactured components.

Photography:

Gamma and contrast transformations can improve brightness and contrast in photographs.

Document Processing:

Gray-level enhancement can improve the visibility of text and characters before OCR.

Security and Surveillance:

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Dark surveillance images can be enhanced to make objects and structures more visible.

Microscopic Imaging:

Low-contrast microscopic images can be enhanced to reveal cellular structures.

Remote Sensing:

Intensity transformations can improve interpretation of images acquired under different illumination conditions

Final Conclusion

The experiment successfully demonstrated the use of different gray-level transformation techniques for enhancing five degraded images. The images provided in the assignment exhibited different visual characteristics, including low contrast, dark intensity levels, and limited visibility of important structures. Contrast stretching was useful for images having a limited range of gray levels because it expanded the available intensity range. Logarithmic transformation enhanced information in darker regions by expanding low-intensity values. Power-law/gamma transformation provided flexible brightness control and was particularly useful for very dark images when $\gamma < 1$.

The experiment also demonstrated that there is no single gray-level transformation that is optimal for every degraded image. The selection of the transformation and its parameters depends on the characteristics of the input image.

The techniques improved the visual appearance of the degraded images, but they also have limitations. They cannot recover information that was completely lost during image acquisition and may sometimes amplify noise or unwanted variations. Therefore, for severely degraded images, gray-level transformations may need to be combined with other techniques such as spatial filtering, noise reduction, histogram-based methods, or image restoration.

Overall, the experiment provided practical understanding of how gray-level transformations can be selected and applied to improve the visual quality and interpretability of degraded digital images.

GitHub:

https://github.com/abhinayreddy-7/DSIP